Srikrishnaa **Vadivel** | Embedded Systems Engineer | FPGA | Robotics | Sensor Logging

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Profile

Embedded systems engineer with field-tested sensor logging, MCU firmware, FPGA logic, robotics, and scientific instrumentation experience. Comfortable from oscilloscope validation and low-level C through host-side visualization and data systems.

Experience

Probe Technologies / Weatherford
Software and Embedded Engineer *2019–2021*

- Programmed C8051-class microcontrollers and Lattice FPGA logic for sensor logging in geological well tools.
- Validated embedded systems with oscilloscopes, bench instrumentation, and field deployments.
- Built host-side CUDA/OpenGL software for large acoustic waveform processing and visualization.

Yale University
Embedded Systems Teaching Assistant / PhD Researcher *2021–present*

- Served as embedded systems TA while pursuing PhD research in computational materials and scientific computing.
- Used Linux, C/C++, Python, MPI/OpenMP, PETSc/SLEPc, and HPC systems for simulation workflows.

Cornell ECE
Robotics Hand Project *2017*

- Built an embedded robotic hand project later published in Circuit Cellar.

Portfolio Projects

MIPS Processor: Verilog single-cycle CPU and self-checking Icarus Verilog test bench.

Matrix-Multiply Accelerator: Small AI accelerator block with start/done handshake and test bench.

ROS 2 TurtleBot Simulation: Robotics simulation plan with trajectory analysis artifacts.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BA, Electrical and Computer Engineering *2013–2017*

Skills

Programming: C, C++, Python, Git

Hardware: Verilog/SystemVerilog, FPGA, MCU firmware, ADC logging
Interfaces: SPI, I2C, UART
Validation: Oscilloscopes, bench instrumentation
Systems: Linux, CUDA, OpenGL, FreeRTOS concepts